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Overcoming Data Imbalance in Molecular Profiling: A Machine Learning Approach for Drug Prediction

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Abstract

Drug discovery is increasingly driven by data-centric approaches, yet the pervasive issue of class imbalance in molecular datasets poses a critical barrier to robust predictive modeling. This study introduces a machine learning (ML)-powered framework specifically designed to counter class imbalance while predicting disease-specific molecular activity. Integrating a triad of molecular descriptors—RDKit, Extended Connectivity Fingerprints (ECFP), and MACCS keys—the framework captures structural and physicochemical diversity for comprehensive molecular representation. To mitigate imbalance, a hybrid strategy combining KMeans-based undersampling of majority classes with dynamic class-weighting is employed, preserving data diversity while enhancing minority class sensitivity. Four ML algorithms—Random Forest, XGBoost, LightGBM, and Support Vector Machines (SVM)—are rigorously benchmarked using metrics such as F1 score, ROC AUC, precision, recall, and accuracy. Among them, SVM paired with ECFP exhibits optimal performance, offering a strong trade-off between specificity and sensitivity. The pipeline also demonstrates adaptability to SMILES-based input, noise resilience, and scalability, making it well-suited for early-stage screening. By foregrounding imbalance-aware learning, this work advances a scalable, interpretable, and cost-efficient approach for bioactivity prediction, contributing to more equitable and effective drug discovery pipelines.

Keywords: Molecular Representations, RDKit Descriptors, Extended Connectivity Fingerprints, Bioactivity Classification, Machine Learning, Drug Discovery, Imbalanced Dataset

1 Introduction

Drug discovery is a costly and time-intensive process, with each development stage demanding precise optimization of molecular properties to ensure safety and efficacy. It typically spans 10–13 years and incurs costs ranging from 1 to 2 billion USD per approved drug [1, 2]. Despite these investments, high attrition rates in clinical trials—where over two-thirds of time and resources are expended—are often linked to insufficient preclinical screening of crucial molecular attributes like toxicity, solubility, and bioactivity [3, 4].

To mitigate these inefficiencies, computer-aided drug design (CADD) techniques have become increasingly prominent [5]. These approaches use computational models to predict molecular behavior, reducing experimental burdens. Traditional methods like molecular docking [6], molecular dynamics simulations [7], and scoring functions [8] have delivered valuable insights but often suffer from computational intensity or limited predictive power [6, 8].

Machine learning (ML) now stands at the forefront of predictive modeling in drug discovery, capable of capturing intricate, non-linear relationships in chemical data [9]. With molecular representations such as SMILES strings and descriptors including RDKit, ECFP, and MACCS keys, ML facilitates the efficient prediction of physicochemical and biological properties [10]. However, when applied to disease-specific activity prediction, ML faces a major hurdle: class imbalance. Most bioactivity datasets contain far more inactive than active compounds, resulting in biased learning and reduced generalizability [11, 12].

Addressing this imbalance is essential for reliable disease-targeted modeling. Prior work has explored methods such as data resampling, synthetic minority oversampling, and cost-sensitive learning [13, 14]. Our study extends this field by employing KMeans clustering-based undersampling, which maintains the structural diversity of majority classes, alongside dynamic class-weighting to fine-tune model sensitivity. This dual-pronged strategy directly counters the distortion caused by imbalanced data, leading to improved minority class recall without compromising overall model performance.

We propose a novel ML framework that integrates molecular featurizers and imbalance-aware learning strategies to predict disease-specific bioactivity. Descriptors such as RDKit, ECFP, and MACCS keys are used to extract rich molecular features from SMILES strings. ML models including Random Forest, XGBoost, LightGBM, and SVM are benchmarked for their performance across balanced metrics. Unlike prior work focused on generic property prediction, our pipeline prioritizes disease-relevance, offering high sensitivity even in skewed datasets.

Key innovations of the framework include:

- **Descriptor Diversity:** Combining multiple molecular featurizers allows for a more nuanced representation of chemical space.
- **Clustering-Based Undersampling:** KMeans clustering reduces majority class size while preserving structural variance, improving model learning.
- **Class-Weighted Learning:** Adaptive class-weighting boosts sensitivity to minority classes, mitigating the bias introduced by dataset skewness.

By aligning these techniques with SMILES-based datasets, the framework supports early-stage, scalable, and cost-effective screening of potential bioactives. This work contributes a robust, interpretable, and imbalance-aware solution for advancing predictive drug discovery [15, 16].

2 Related Study

The integration of molecular representations with machine learning (ML) algorithms has significantly transformed computational drug discovery, addressing challenges in modeling diverse molecular properties and bioactivity classifications. By reducing the reliance on experimental validation, these approaches accelerate tasks such as solubility prediction, ADMET (absorption, distribution, metabolism, excretion, and toxicity) modeling, and ligand-binding activity prediction. This section elaborates on molecular representations, ML methodologies, and their mathematical underpinnings, emphasizing their contributions to molecular property prediction and classification.

2.1 Molecular Representations: Descriptors and Fingerprints

Molecular representations are the foundation of cheminformatics—encoding structural, physicochemical, and electronic properties into machine-readable formats. They facilitate tasks such as quantitative structure-activity relationships (QSAR), toxicity modeling, ligand-receptor interaction prediction, and molecular classification.

Extended Connectivity Fingerprints (ECFP)

ECFP generates fixed-length binary vectors through iterative hashing of atomic neighborhoods. The mathematical formulation involves:

$$v_i = h(v_{i-1}, \text{neighbors}(v_{i-1})), \quad (1)$$

where v_i is the hashed vector at iteration i , and h is a hashing function. ECFP retains critical substructural features that influence molecular activity, making it an ideal choice for structure-activity relationship (SAR) modeling [10]. Its robustness and topological detail enable superior performance in ligand-based virtual screening and classification tasks such as toxicity and binding affinity prediction [14].

MACCS Keys

MACCS keys encode 166 predefined molecular substructures as binary vectors. These representations are computationally efficient, making them suitable for scaffold analysis and similarity-based screening [17]. Applications span toxicity prediction, environmental risk assessments, and classification of molecular subtypes based on bioactivity.

RDKit Descriptors

RDKit descriptors encode molecular properties such as molecular weight (MW), logP (partition coefficient), hydrogen bond donors (HBD), and polar surface area (PSA).

These descriptors are scalar or vector-based and serve as quantitative predictors in QSAR modeling. For instance, logP, which measures lipophilicity, is computed as:

$$\log P = \sum_{i=1}^N f(\text{atom}_i), \quad (2)$$

where $f(\text{atom}_i)$ denotes the hydrophobic contribution of atom i [12]. RDKit descriptors have been widely applied in solubility modeling [18], hepatotoxicity prediction [8], and environmental toxicity studies [11]. Their applications extend to classifying molecules into active and inactive categories for bioactivity studies. To mitigate redundancy and dimensionality issues, feature selection techniques such as Recursive Feature Elimination (RFE), Principal Component Analysis (PCA), or Lasso regularization are often employed.

2.2 Machine Learning Algorithms for Molecular Property Prediction and Classification

ML algorithms establish nonlinear relationships between molecular representations and target properties, enabling high-accuracy predictions in cheminformatics [19]. Their mathematical flexibility and scalability make them indispensable for modeling diverse molecular properties and classifying molecules into activity or toxicity categories.

Gradient Boosting (XGBoost, LightGBM)

Gradient Boosting sequentially minimizes a loss function L , optimizing weak learners (trees) iteratively:

$$L = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (3)$$

where y_i is the observed value and $\hat{y}_i$ is the predicted value. XGBoost and LightGBM introduce enhancements like regularization, class weighting, and leaf-wise growth, making them effective for imbalanced datasets [12]. These models are extensively used for molecular property prediction and classification tasks, such as identifying high-affinity ligands and predicting molecular reactivity.

Random Forest (RF)

Random Forest constructs an ensemble of decision trees, each trained on random subsets of features and samples. Impurity measures such as Gini Index or Entropy are minimized at each node:

$$I = 1 - \sum_{j=1}^k p_j^2 \quad (\text{Gini Index}), \quad \text{or} \quad I = - \sum_{j=1}^k p_j \log p_j \quad (\text{Entropy}), \quad (4)$$

where p_j represents the proportion of samples in class j . RF excels in ADMET predictions, bioactivity classification, and solubility modeling [5, 20], with built-in feature

importance scoring for interpretable models. These properties make it an effective algorithm for classifying active and inactive compounds or categorizing toxicity levels.

Support Vector Machines (SVM)

SVM finds the optimal hyperplane that separates classes by maximizing the margin between data points and the decision boundary. The margin is defined as:

$$\text{Margin} = \frac{2}{\|\mathbf{w}\|}, \quad (5)$$

where $\mathbf{w}$ is the weight vector. Kernel functions extend SVM to non-linear decision boundaries, such as the Radial Basis Function (RBF):

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2). \quad (6)$$

SVM is particularly effective for high-dimensional datasets, demonstrating superior performance in ligand-receptor interaction prediction and multi-class molecular classification for toxicity and environmental risk assessments [13, 17].

Innovations in the Proposed Pipeline

The proposed pipeline leverages clustering-based undersampling techniques to address dataset imbalances while preserving molecular diversity. By integrating RDKit descriptors, ECFP, and MACCS keys with ensemble ML models like RF, XGBoost, and LightGBM, it achieves enhanced classification performance for diverse molecular datasets. Feature selection further improves interpretability, making it scalable and effective for small yet diverse datasets.

Advances in Clustering for Imbalanced Data

Clustering techniques like KMeans address class imbalance by undersampling majority classes while preserving structural diversity [21]. When integrated with descriptors, clustering improves ML model performance by enhancing data representativeness.

Why Machine Learning Over Deep Learning?

While deep learning excels in large datasets, machine learning remains preferred in cheminformatics due to its data efficiency, computational scalability, and interpretability. RF, XGBoost, and SVM offer robust performance on small datasets common in early-stage drug discovery. Their ability to rank feature importance makes them indispensable for deriving insights from molecular representations [5, 19].

3 Proposed Pipeline

The proposed pipeline integrates advanced molecular representations, innovative class balancing techniques, and state-of-the-art machine learning models to address the

inherent challenges of a diverse and imbalanced *SARS-CoV-2* dataset, samples of which are shown in Figure 1. This structured pipeline, summarized in a block diagram shown in Figure 2, comprises several key stages, described in detail below.

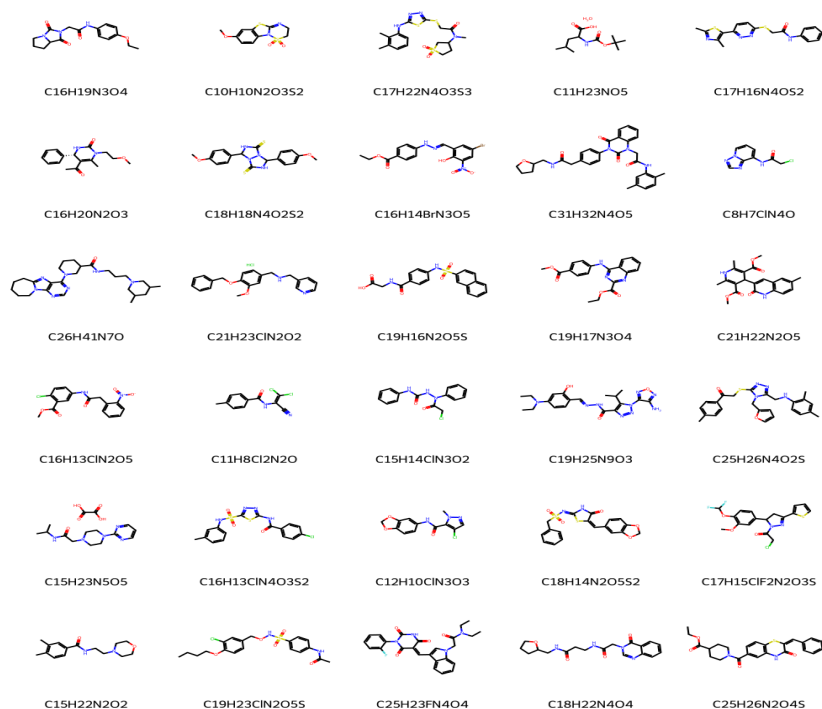


Fig. 1 Sample representation of SARS-CoV-2 Dataset

The process begins with **data preprocessing**, where the dataset, represented as **SMILES (Simplified Molecular Input Line Entry System) strings**, undergoes filtering to ensure relevance and validity. Only molecules with a minimum of 12 atoms are retained, excluding smaller, less meaningful structures. Invalid SMILES strings, which cannot be translated into valid molecular representations using RDKit, are removed during the filtering process. This ensures that only chemically valid molecules proceed to subsequent steps.

To address the imbalance between inactive and active compounds, the *class balancing* stage employs KMeans clustering on the majority class using a targeted subset of RDKit descriptors—molecular weight, logP, hydrogen bond donors, hydrogen bond acceptors, and aromatic ring count. This clustering groups structurally similar molecules, enabling representative sampling from each cluster to ensure diversity and proportional representation within the majority class. The minority class, containing active compounds, is retained in its entirety to preserve crucial bioactivity information. By combining these balanced classes, the dataset is split into training (70%) and testing (30%) subsets, ensuring unbiased evaluation and avoiding data leakage.

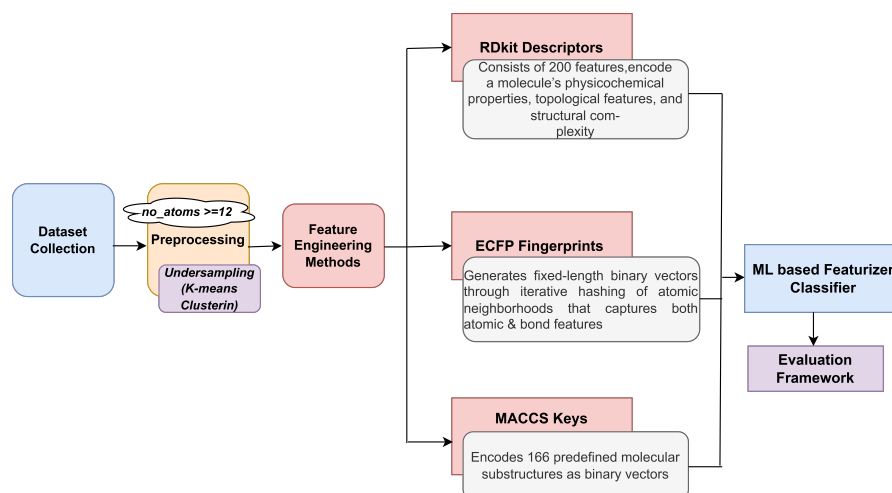


Fig. 2 Proposed ML-Featurizer Pipeline for Molecular Activity Classifier

In the **feature engineering** stage, molecular representations are derived to capture diverse and complementary characteristics of the molecules. Three primary representations are utilized: RDKit descriptors, Extended Connectivity Fingerprints (ECFP), and MACCS keys. RDKit descriptors, comprising over 200 features, encode a molecule's physicochemical properties, topological features, and structural complexity, providing a holistic molecular characterization. ECFP, computed using the Morgan algorithm, captures substructural motifs by encoding atom-level connectivity and relationships within the molecular graph, effectively linking structure to potential bioactivity. MACCS keys, a predefined set of 167 binary features, represent chemical substructures in a compact and interpretable format, ensuring computational efficiency for downstream tasks.

Within this stage, *data integrity and noise reduction* ensures clean datasets for reliable analysis. Rows containing NaN values in feature matrices, which may arise from invalid or missing features during calculation, are identified and removed. This step prevents errors during model training and ensures robust predictions. Both probabilistic and categorical predictions are subsequently analyzed to validate data quality and model readiness.

The **machine learning models** stage integrates dynamic class weights derived from the training data, ensuring fair treatment of both classes during model training. Four machine learning algorithms are utilized to predict molecular activity. **Random Forest**, an ensemble-based method, constructs multiple decision trees and incorporates class weighting to handle imbalanced data effectively. XGBoost, known for its efficiency, employs the `scale_pos_weight` parameter to dynamically adjust for class imbalance during gradient-boosted tree training. LightGBM, another gradient-boosted algorithm, applies automatic class weighting for fairness and computational efficiency. Support Vector Machines (SVM), leveraging weighted loss functions, optimizes the separation of classes in imbalanced datasets by focusing on correctly identifying

the minority class. These models are systematically evaluated with each molecular representation to explore the interplay between feature diversity and algorithmic adaptability.

The **evaluation framework** assesses the performance of these models using comprehensive metrics. Accuracy measures the overall correctness of predictions, while the F1 score balances precision and recall to evaluate effectiveness on imbalanced datasets. ROC AUC evaluates the trade-off between sensitivity and specificity across varying thresholds, offering a holistic view of model discriminative power. Precision and recall provide additional insights into the model's ability to correctly identify active compounds while minimizing false positives.

This robust pipeline establishes a scientifically rigorous and adaptable framework for SARS-CoV-2 molecular classification. By addressing key challenges such as dataset imbalance, molecular diversity, and data noise, it provides a benchmark for predictive modeling in virology, drug discovery, and cheminformatics.

4 Results and Discussion

The performance of the proposed pipeline was rigorously evaluated using machine learning models—Random Forest, XGBoost, Support Vector Machine (SVM), and LightGBM—integrated with molecular representations (RDKit descriptors, Extended Connectivity Fingerprints (ECFP), and MACCS keys). Evaluation metrics such as accuracy, F1 score, ROC AUC, precision, and recall were utilized to assess the models' capability to classify SARS-CoV-2 molecular activity. A detailed summary of these metrics is provided in Table 1, while Figures 3 and 4 present visual representations of the comparative performance across the models and molecular representations.

Table 1 Performance Metrics for Machine Learning Models with Different Molecular Representations. SVM with ECFP demonstrates the highest overall performance, validating its efficacy for bioactivity classification in cheminformatics.

Model (Representation)	Accuracy	F1 Score	ROC AUC	Precision	Recall
Random Forest (RDKit)	0.881243	0.840417	0.792454	0.823529	0.118644
XGBoost (RDKit)	0.876804	0.862532	0.775602	0.550725	0.322034
SVM (RDKit)	0.134295	0.037055	0.470236	0.131403	1.0
LightGBM (RDKit)	0.873474	0.855120	0.778687	0.533333	0.271186
Random Forest (ECFP)	0.891232	0.861789	0.790349	0.833333	0.211864
XGBoost (ECFP)	0.860155	0.856941	0.755504	0.462264	0.415254
SVM (ECFP)	0.897891	0.887587	0.796567	0.675676	0.423729
LightGBM (ECFP)	0.863485	0.858327	0.767020	0.474747	0.398305
Random Forest (MACCS)	0.883463	0.847536	0.768156	0.782609	0.152542
XGBoost (MACCS)	0.862375	0.851535	0.748523	0.463415	0.322034
SVM (MACCS)	0.764706	0.794599	0.771728	0.302521	0.610169
LightGBM (MACCS)	0.841287	0.842131	0.744421	0.396694	0.406780

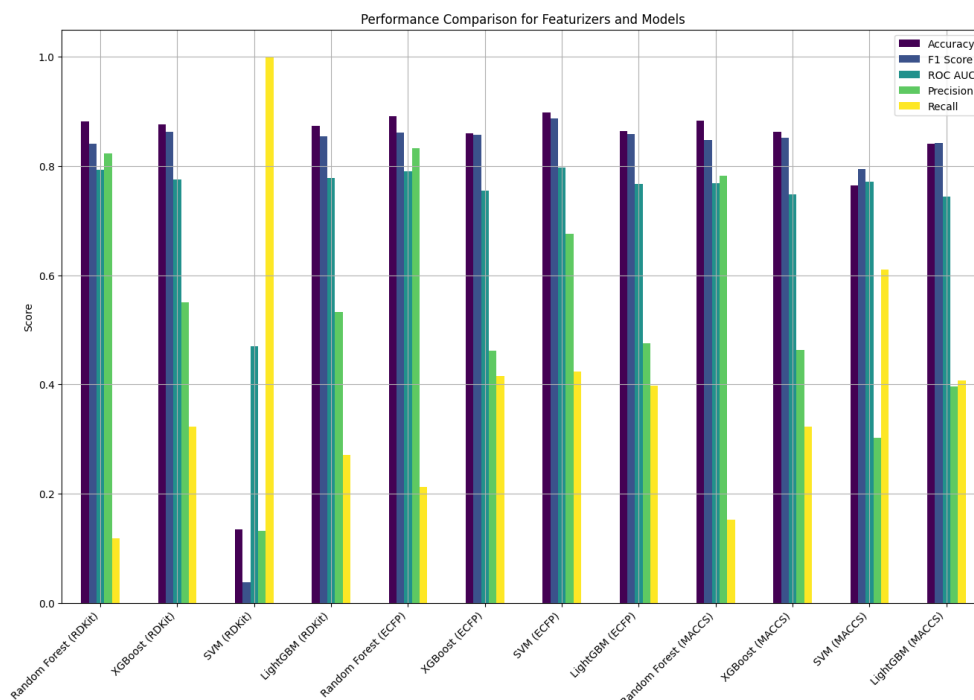


Fig. 3 Performance Comparison Across Machine Learning Models and Molecular Representations. SVM with ECFP outperformed other combinations

4.1 Performance Insights

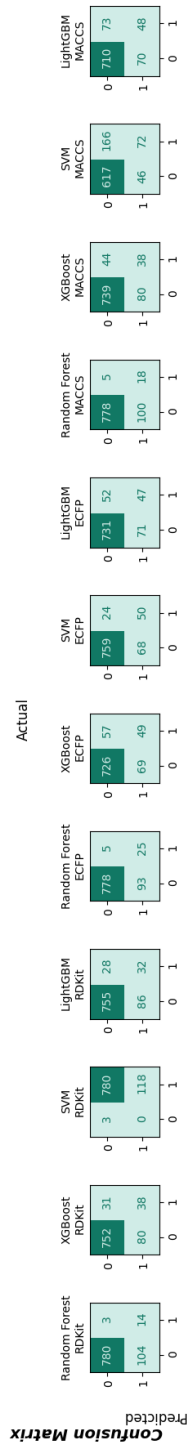
Model-Representation Performance

Among the evaluated combinations, **SVM with ECFP** demonstrated the highest overall performance, achieving an accuracy of 0.897891, F1 score of 0.887587, and ROC AUC of 0.796567. This combination effectively balanced sensitivity and specificity, with precision and recall scores of 0.675676 and 0.423729, respectively. The results validate the use of ECFP, which captures topological and substructural molecular features essential for accurate classification in cheminformatics.

Impact of Molecular Representations

ECFP consistently produced superior performance metrics across all models, underscoring its robustness in encoding molecular features. RDKit descriptors showed competitive results, particularly when combined with LightGBM, achieving an accuracy of 0.873474. However, MACCS keys underperformed relative to the other representations, likely due to their predefined structure, which limits flexibility in capturing intricate molecular patterns.

Performance Metrics Across Models and Representations



False Positive Rate

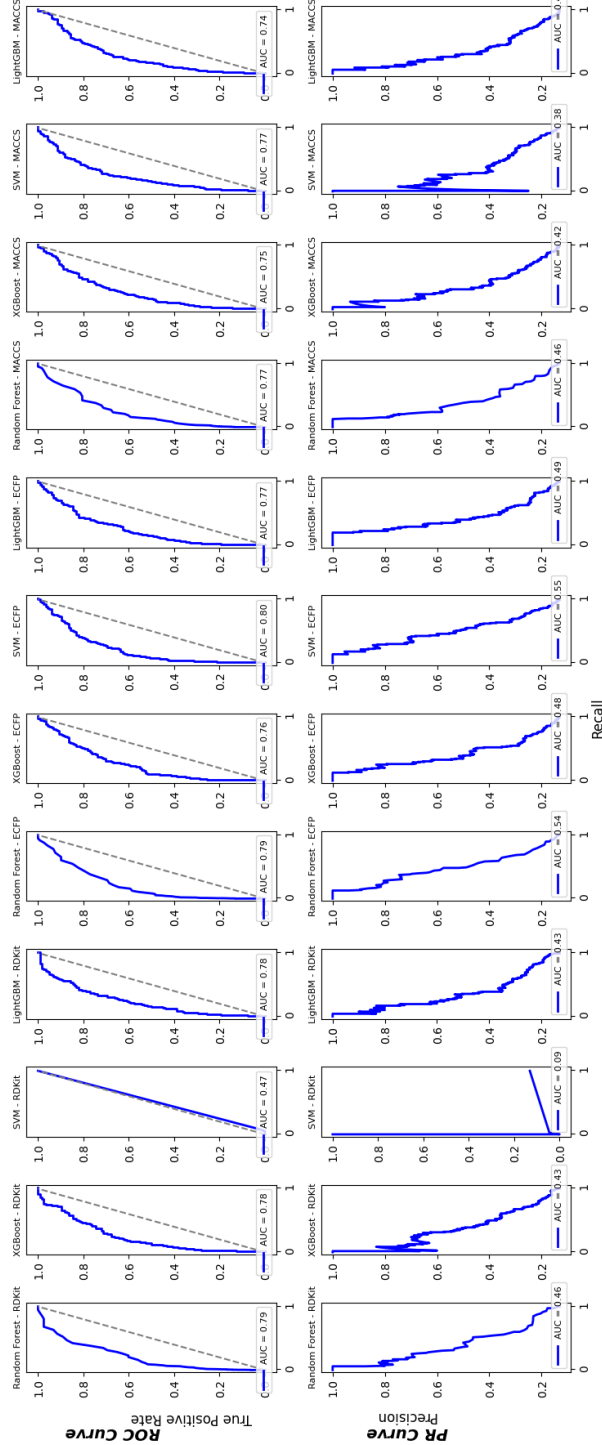


Fig. 4. Performance Metrics Across Machine Learning Models and Representations. Top row: Confusion matrices; Middle row: ROC curves; Bottom row: Precision-Recall (PR) curves. SVM with ECFP demonstrates balanced performance across all metrics.

Model Efficiency

LightGBM with RDKit and ECFP demonstrated robust and balanced performance, making it suitable for large-scale bioactivity classification tasks. Conversely, SVM with RDKit exhibited the weakest performance (accuracy: 0.134295, F1 score: 0.037055), indicating its sensitivity to feature representation and the need for richer descriptors.

4.2 Visualization Analysis

Figure 3 provides a bar-chart comparison of model performance across the representations, revealing that **SVM with ECFP** consistently outperformed other combinations. Figure 4 elaborates these findings:

- **Confusion Matrices:** SVM with ECFP displayed the most balanced classification, minimizing both false positives and false negatives. In contrast, SVM with RDKit showed significant misclassifications.
- **ROC and PR Curves:** The ROC and PR curves for SVM with ECFP highlight its superior discriminative capability, with consistently higher AUC values compared to other combinations.

These visual analyses reaffirm that the proposed pipeline effectively integrates molecular representations and machine learning models to address the challenges of imbalanced datasets in cheminformatics.

5 Conclusion

This study presented a machine learning-driven pipeline for molecular activity classification, focusing on SARS-CoV-2 datasets. The pipeline integrates advanced molecular representations (RDKit descriptors, ECFP, and MACCS keys) with clustering-based class balancing techniques and state-of-the-art machine learning models. Among the evaluated combinations, **SVM with ECFP** emerged as the most effective, achieving superior performance metrics across all evaluation criteria.

The results emphasize the importance of leveraging diverse molecular representations for bioactivity classification. While ECFP proved to be the most robust representation, RDKit descriptors also demonstrated strong potential, especially when paired with LightGBM. The underperformance of MACCS keys highlights the need for flexible and detailed molecular encodings. Additionally, the results demonstrate the practicality and interpretability of machine learning approaches in cheminformatics, especially for imbalanced datasets where deep learning methods may struggle due to high data requirements and computational complexity.

Future work can focus on hybrid frameworks that integrate the interpretability of machine learning models with the feature extraction capabilities of deep learning. Such an approach could further enhance scalability and accuracy for molecular classification tasks.

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